Analytical Chemistry 1 (125101)

Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

Framework of this course

❖ Introduction (L1)

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

2.6 gravimetric analysis (L12)

❖ Summary (L13)

2.3 Potentiometry in titration

2.3 Potentiometry in Titrations

Overview

2.3-1 Galvanic cell

2.3-2 Standard reduction potential (E^o)

2.3-3 Nernst equation

2.3-4 Electrodes types

❑ Electroanalysis

Electroanalytical methods are based on redox reactions performed through an electrical circuit.

❑ Redox reactions

A reaction is which an electron is transferred between atoms/ions,
The reaction is based on differences in the electron affinity

Oxidant Reductant

$$\mathbf{Fe^{3+} + V^{2+} \rightarrow Fe^{2+} + V^{3+}}$$

Vanadium

Fe^{3+} takes an electron; V^{2+} donates an electron

Oxidation: loss of electrons, oxidation state↑ $V^{2+} \rightarrow V^{3+} + e$
Reduction: gain of electrons, oxidation state ↓ $Fe^{3+} + e \rightarrow Fe^{2+}$

2.3-1 Galvanic cell

- Utilizes a spontaneous chemical reaction to produce electricity.
- One component is oxidized and the other is reduced.

Oxidation: $Cd(s) \rightleftharpoons Cd^{2+}(aq) + 2e^-$ Cadmium

Reduction: $2AgCl(s) + 2e^- \rightleftharpoons 2Ag(s) + 2Cl^-(aq)$

Net reaction: $Cd(s) + 2AgCl(s) \rightleftharpoons Cd^{2+}(aq) + 2Ag(s) + 2Cl^-(aq)$

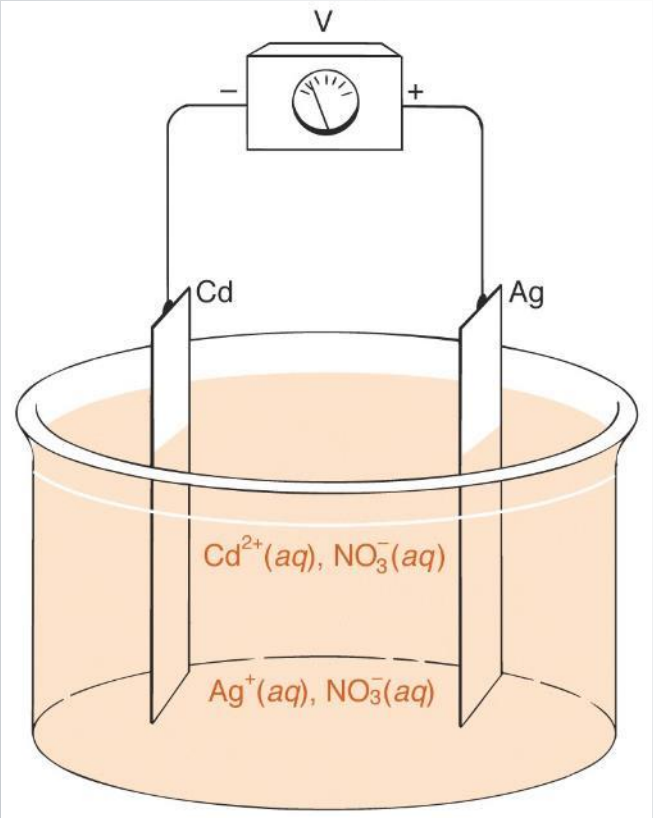
The two components have to be separated so that the electrons must pass through the electrical circuit.

Let's construct a cell for the following reactions:

Anode: $Cd(s) \rightleftharpoons Cd^{2+}(aq) + 2e^-$

Cathode: $2Ag^+(aq) + 2e^- \rightleftharpoons 2Ag(s)$

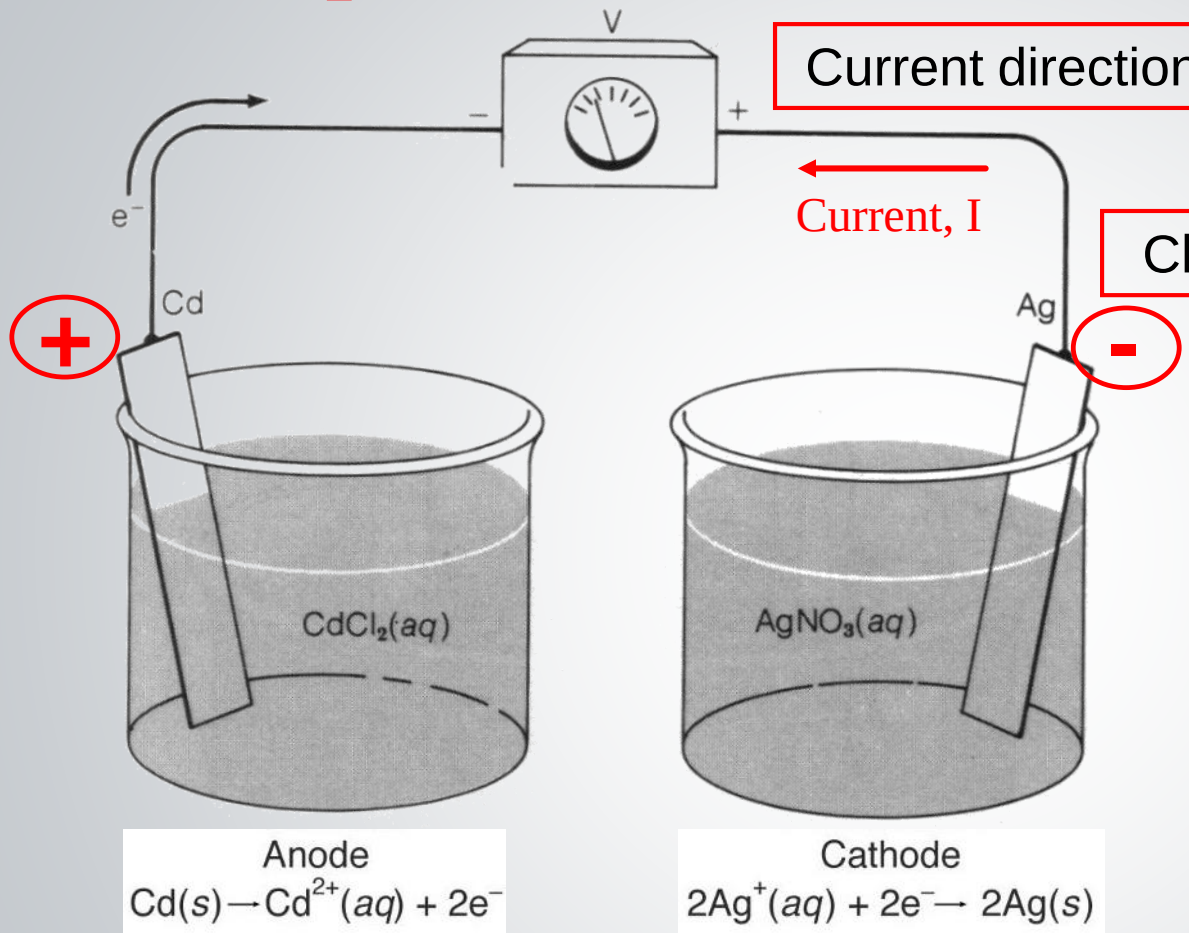
Net reaction: $Cd(s) + 2Ag^+(aq) \rightleftharpoons Cd^{2+}(aq) + 2Ag(s)$



Is current passing?

Almost no current, since Ag^+ in the solution directly reacts on the Cadmium (Cd) electrode surface.

Let's separate the reactions:

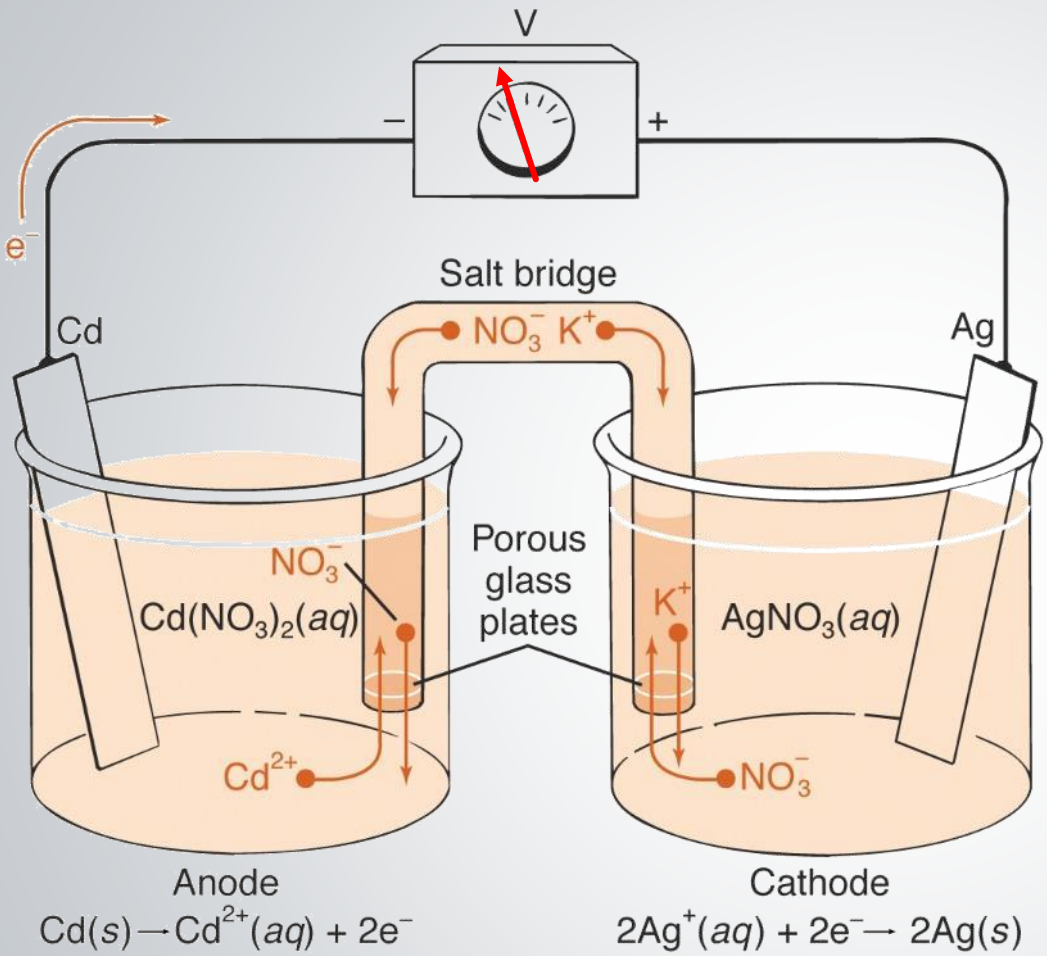


Short-time current

The right cell becomes negative and the left cell becomes positive.

Further electrons are repulsed /prevented and current stops. The current stops when the electrostatic repulsion is equal to the electrochemical force.

Let's add a salt bridge



- **Salt bridge:**

Connect two half-cells. Ions from the salt bridge migrate to the cells and neutralize the charges, which will result in continuous current in the electrical circuit.

Salt bridge: video

Notation: $\mathbf{Cd_{(s)} \mid Cd(NO_3)_{2(aq)} \parallel AgNO_{3(aq)} \mid Ag(s)}$

phase interface salt bridge phase interface

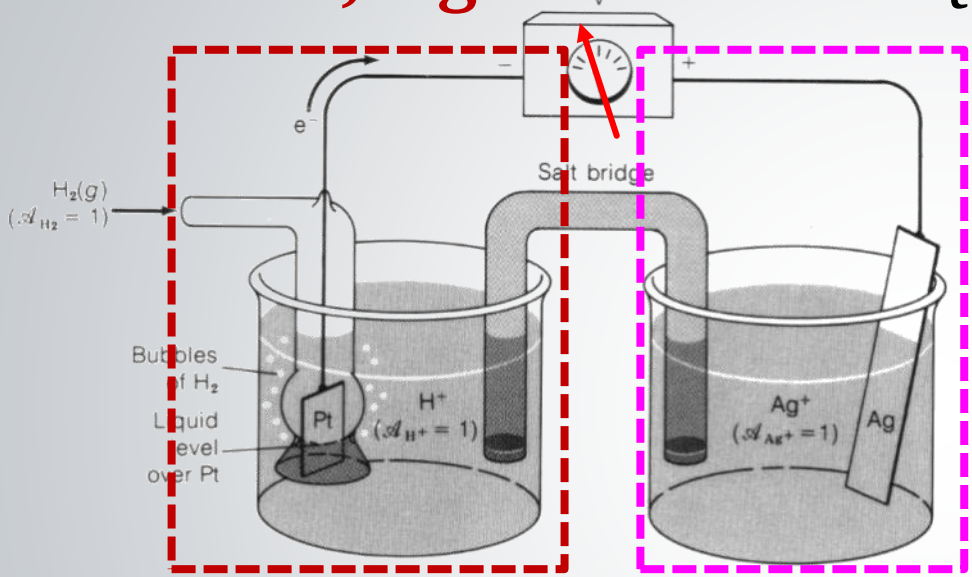
2.3-2 Standard reduction potentials (E^o)

The potential measured in the previous cell is different between the Ag electrode and Cd electrode

We place at one side a standard half-cell so that we can compare the potentials of various reactions in the other half-cell. (Which means potential is a relative concept, like height)

The standard reference cell is assigned to the **Left side, e.g. S.H.E.**

The measured potential for $Ag^+ | Ag$ is assigned to the **right side**



$Pt(s)|H_2(g, \mathscr{A} = 1)|H^+(aq, \mathscr{A} = 1)\|Ag^+(aq, \mathscr{A} = 1)|Ag(s)$

$\frac{1}{2}H_2(g, \mathscr{A} = 1) \rightleftharpoons H^+(aq, \mathscr{A} = 1) + e^-$ $\quad$ $Ag^+ + e^- \rightleftharpoons Ag(s)$

standard hydrogen electrode (S.H.E.).

By convention, $E^\circ = 0$ for S.H.E.

The measured potential

is: **+ 0.799 V**

comparing to standard hydrogen electrode (S.H.E.)

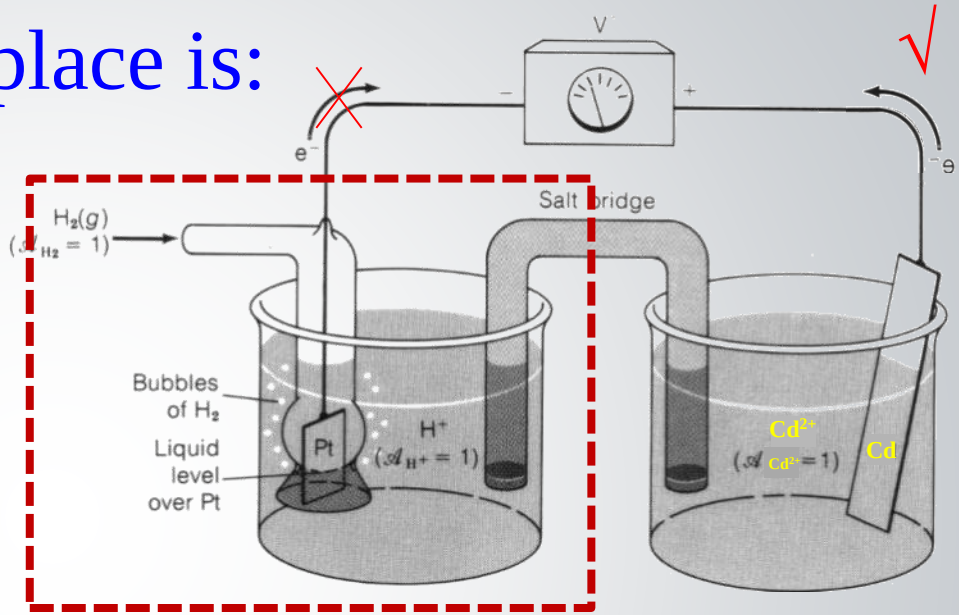
The Standard Reduction reaction on right is:

$$Ag^+ + e^- \rightarrow Ag_{(s)} \quad E^o = 0.799 \text{ V (vs. S.H.E.)}$$

Replacing the right half-cell with Cd/Cd^{2+}

The reaction that takes place is:

$$Cd^{2+} + 2e^- \rightleftharpoons Cd_{(s)}$$



The measured **reduction** potential is: **-0.402 V**

The negative sign indicates that the electrons move to the opposite direction, leaving from the Cd electrode.

The **oxidation** reaction taking place is the opposite reaction:

$$Cd_{(s)} \rightleftharpoons Cd^{2+} + 2e^- \ (E^o = +\ 0.402\ V)$$

Standard reduction potential Table

Ordered redox potentials

Oxidizing agent → Reducing agent	$E°$ (V)
$F_2(g) + 2e^- \rightleftharpoons 2F^-$	2.890
$O_3(g) + 2H^+ + 2e^- \rightleftharpoons O_2(g) + H_2O$	2.075
⋮	
$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$	1.507
⋮	
$Ag^+ + e^- \rightleftharpoons Ag(s)$	0.799
⋮	
$Cu^{2+} + 2e^- \rightleftharpoons Cu(s)$	0.339
⋮	
$2H^+ + 2e^- \rightleftharpoons H_2(g)$	0.000
⋮	
$Cd^{2+} + 2e^- \rightleftharpoons Cd(s)$	−0.402
⋮	
$K^+ + e^- \rightleftharpoons K(s)$	−2.936
$Li^+ + e^- \rightleftharpoons Li(s)$	−3.040

Oxidizing power increases ⇑

Reducing power increases ⇓

★ When you reverse the direction of a reaction, reverse the sign of $E°$

The standard potentials in the Table are against S.H.E., when all conc. are unit 1 M.

2.3-3 Nernst equation

It describes the effect of **conc.** on cell potential.

For the reaction: **aA + bB ⇌ cC + dD**

Nernst eq. provides the cell potential:

$$E = E^{o} - \frac{RT}{nF} \ln \frac{[C]^{c}[D]^{d}}{[A]^{a}[B]^{b}}$$

E^{o} is the cell's standard reduction potential (the potential when all concentrations are 1.0 M)

n is the number of passing electrons in the circuit.

Example:

$$Cd(s) \rightleftharpoons Cd^{2+} + 2e^- \qquad E^\circ = 0.402\ V$$

$$2Ag^+ + 2e^- \rightleftharpoons 2Ag(s) \qquad E^\circ = 0.799\ V$$

$$Cd(s) + 2Ag^+ \rightleftharpoons Cd^{2+} + 2Ag(s) \qquad E^\circ = 1.201\ V$$

$$E = E^\circ - \frac{0.059\,16}{2} \log \frac{[Cd^{2+}]}{[Ag^+]^2}$$

$$\frac{RT}{nF} \ln$$

Converting the natural log (ln) into the base 10 log and T = 298.15 K, R = 8.314 J/(K·mol), Faraday constant F = 9.649×10^4 C/mol (F = $e \cdot N_A$)

2.3-4 Electrode types

- **Reference electrode:**

The half-cell of constant potential (with fixed composition). Used for measuring cell potential

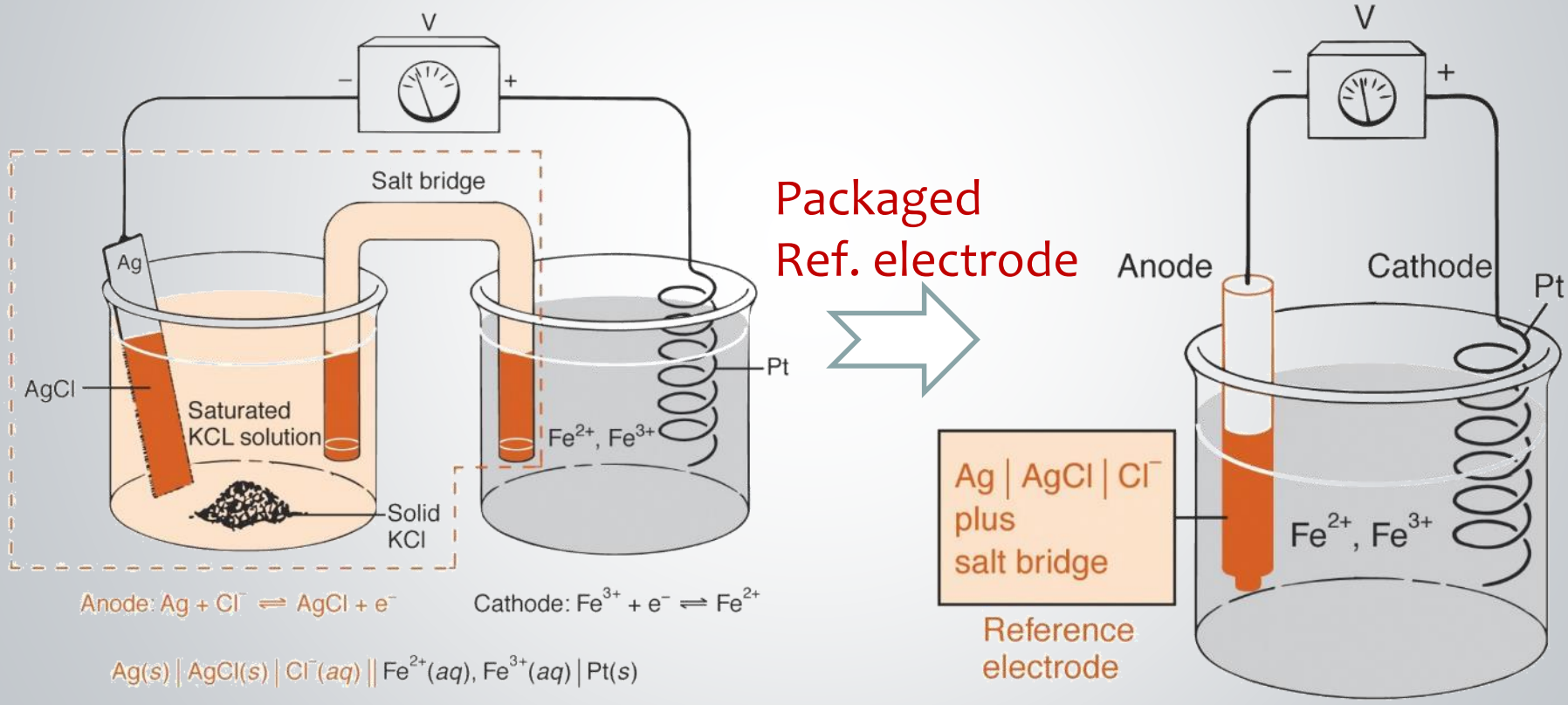
- **Indicator /working electrode:**

Used for passing electrons to/from analyte and responds to concentrations

The whole cell potential is the difference between analyte half-cell (varies with concentration) and the constant potential of the reference electrode. $E = f(C)$

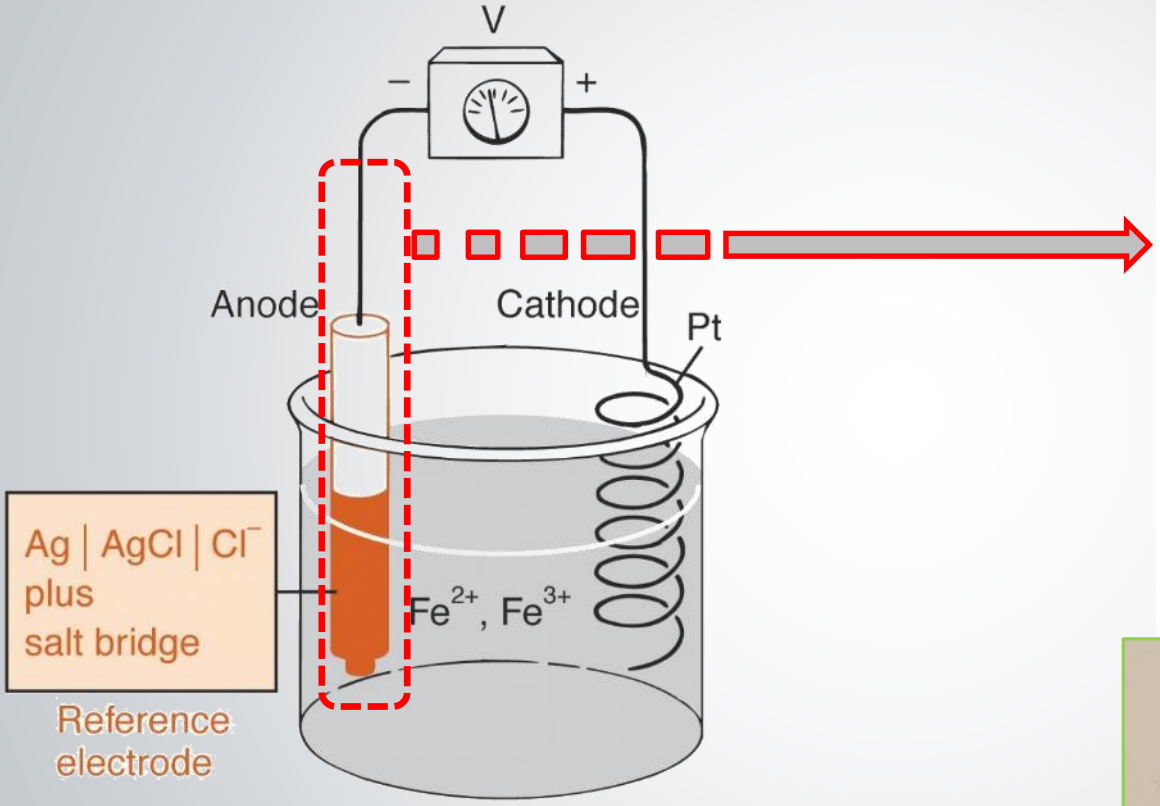
Reference electrode

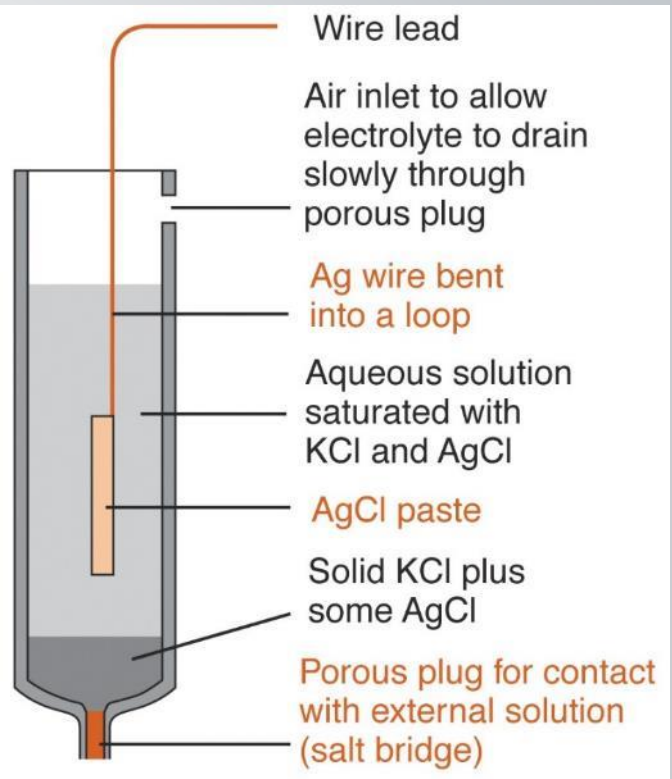
Dip a platinum wire and connect via a salt bridge to a half-cell of **constant** potential of Ag/Ag^+



Cl^- conc. is constant since the solution is saturated with solid KCl

Let's construct a reference electrode based on half-cell of constant potential





Silver-silver chloride reference electrode (available on market)

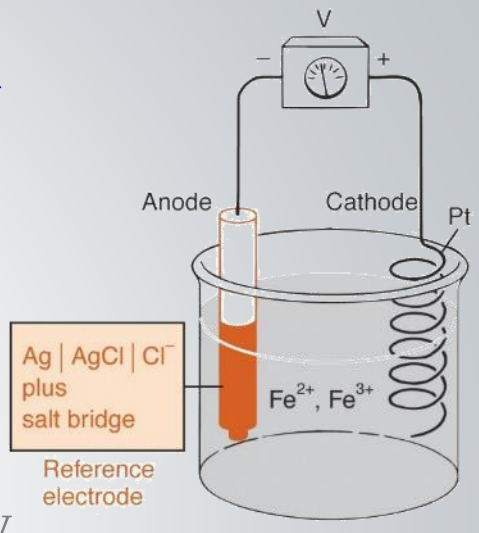
Reference electrode

Example: the conc. of iron ions is unknown

The two half-reactions (written as reductions) are

Right cathode: $Fe^{3+} + e \rightleftharpoons Fe^{2+}$ $E_+^o = 0.767$ V

Left anode: $AgCl\ (s) + e \rightleftharpoons Ag\ (s) + Cl^-$ $E_-^o = 0.222$ V



E^o for each half-cell is the standard reduction potential when the concentration of reactants and products are unity.

Cathode: $E_+ = 0.767 - \frac{0.05916}{1} \log \left(\frac{[Fe2+]}{[Fe3+]}\right)$

Anode: $E_- = 0.222 - \frac{0.05916}{1} \log [Cl^-]$

Keep reference electrode with constant potential

The cell potential is the difference of ($E_+ - E_-$)

$$E\ (cell) = \{0.767 - \frac{0.05916}{1} \log \left(\frac{[Fe2+]}{[Fe3+]}\right)\} - \boxed{\{0.222 - \frac{0.05916}{1} \log [Cl^-]\}}$$

which can be used to measure the quotient of $\frac{[Fe2+]}{[Fe3+]}$

Indicator electrode

Metal indicator electrode:

- Inert electrodes made of Pt, Au, Ag and Cd etc. (transmit *e* only)

- Used for measuring potential resulting from redox reaction on the metal surface.

Ion selective electrode:

- Measures the potential resulting from ions passing through the membrane

Metal indicator electrode

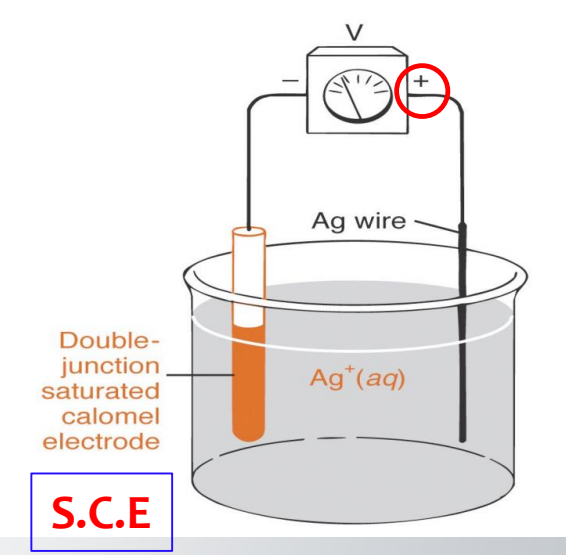
For example: use of Ag and calomel electrodes to measure $[Ag^+]$

Reaction at the Ag indicator electrode:

$Ag^+ + e \rightleftharpoons Ag\ (s) \quad E_+^o = 0.799\ V$

Reference Saturated calomel electrode (S.C.E.):

$HgCl_2\ (s) + 2e \rightleftharpoons Hg\ (l) + 2Cl^- \quad E_- = 0.241\ V$



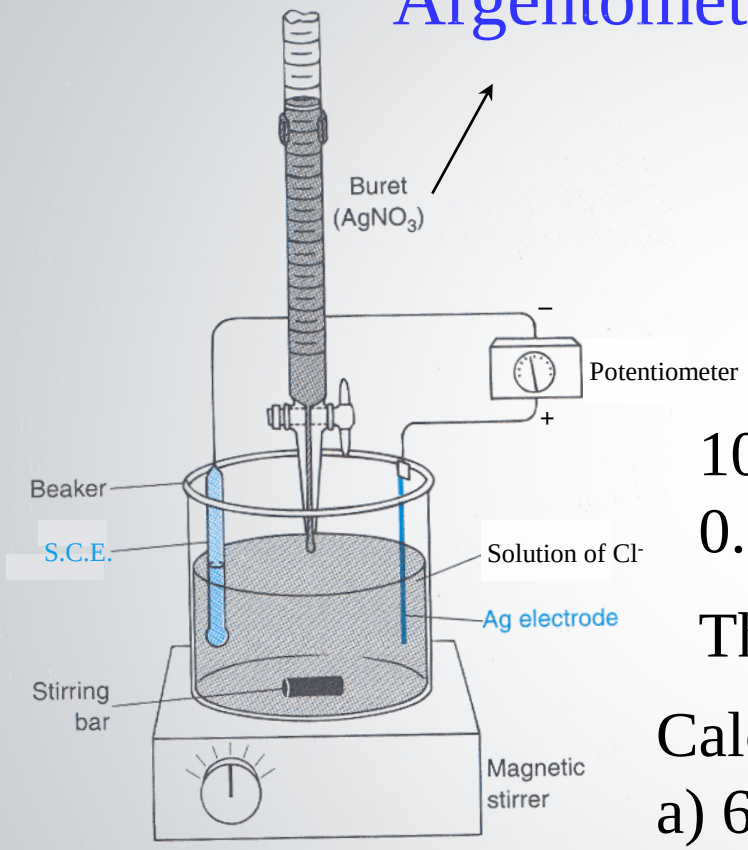
The Nernst equation for the entire cell is:

$$E_{cell} = E_+ - E_- = \underbrace{\left(0.799 - 0.05916 \log \frac{1}{[Ag^+]}\right)}_{\text{Potential of Ag | Ag}^+\text{ indicator electrode}} - \underbrace{0.241}_{\text{Potential of S.C.E Reference electrode}}$$

$$\boxed{E_{cell} = 0.558 + 0.05916 \log [Ag^+]}$$

(a) Potentiometric on precipitation titration

Argentometric titration (Titration with Ag^+)



100 mL 0.1 M NaCl was titrated with 0.1 M $AgNO_3$ ($K_{sp} = 1.8\times10^{-10}$)

The potential of the cell was monitored.

Calculate the potential when addition of a) 65 mL; b) 100 mL; c) 135 mL of $AgNO_3$

Apparatus for measuring titration curves. The silver electrode responds to the change of $[Ag^+]$, S.C.E. provides a reference potential

(a) Potentiometric on precipitation titration

a) At 65 mL, 65% $AgNO_3$ has been precipitated and 35% remains:

$$[Cl^-] = 35\% \times 0.1\ M \times \frac{100\ mL}{100\ mL + 65\ mL} = 0.0212\ M$$

$$[Ag^+] = \frac{Ksp}{[Cl^-]} = 8.5 \times 10^{-9}\ M$$

$$E_+ = 0.558 + 0.05916 \log [Ag^+] = 0.081\ V$$

b) At 100 mL, 100% $AgNO_3$ has been precipitated:

$$[Ag^+] = \sqrt{Ksp} = 1.34 \times 10^{-5}\ M$$

$$E_+ = 0.558 + 0.05916 \log [Ag^+] = 0.27\ V$$

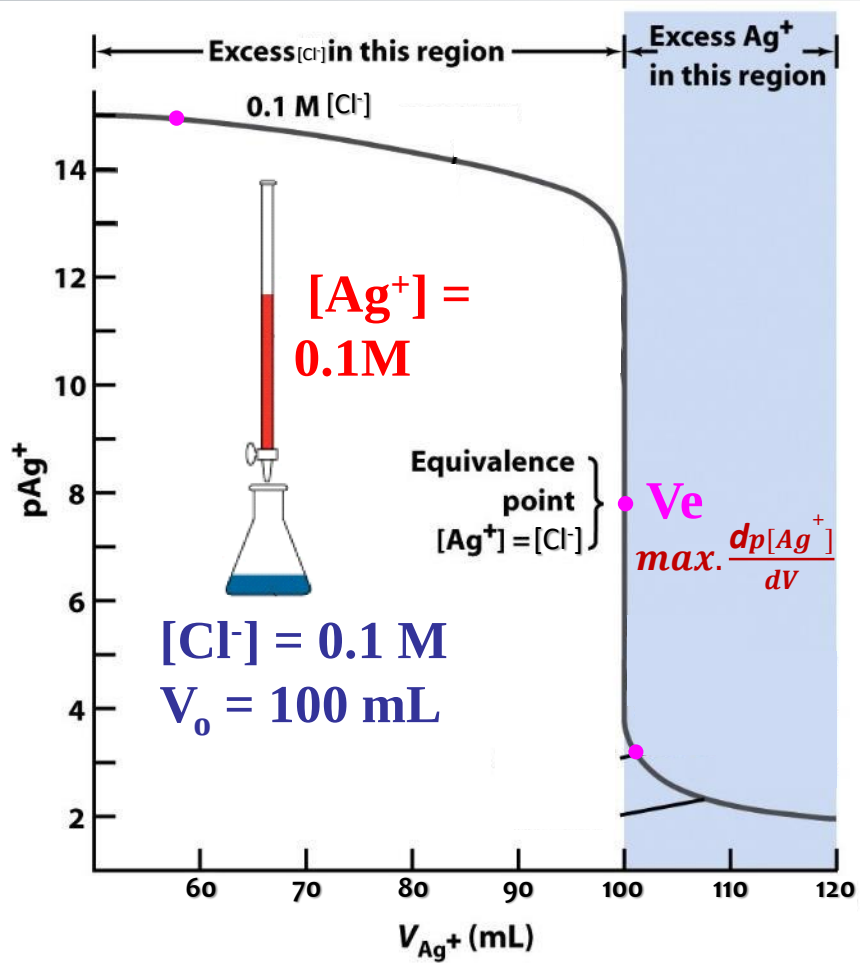
c) At 135 mL, an excess of 35 mL of $AgNO_3$ left:

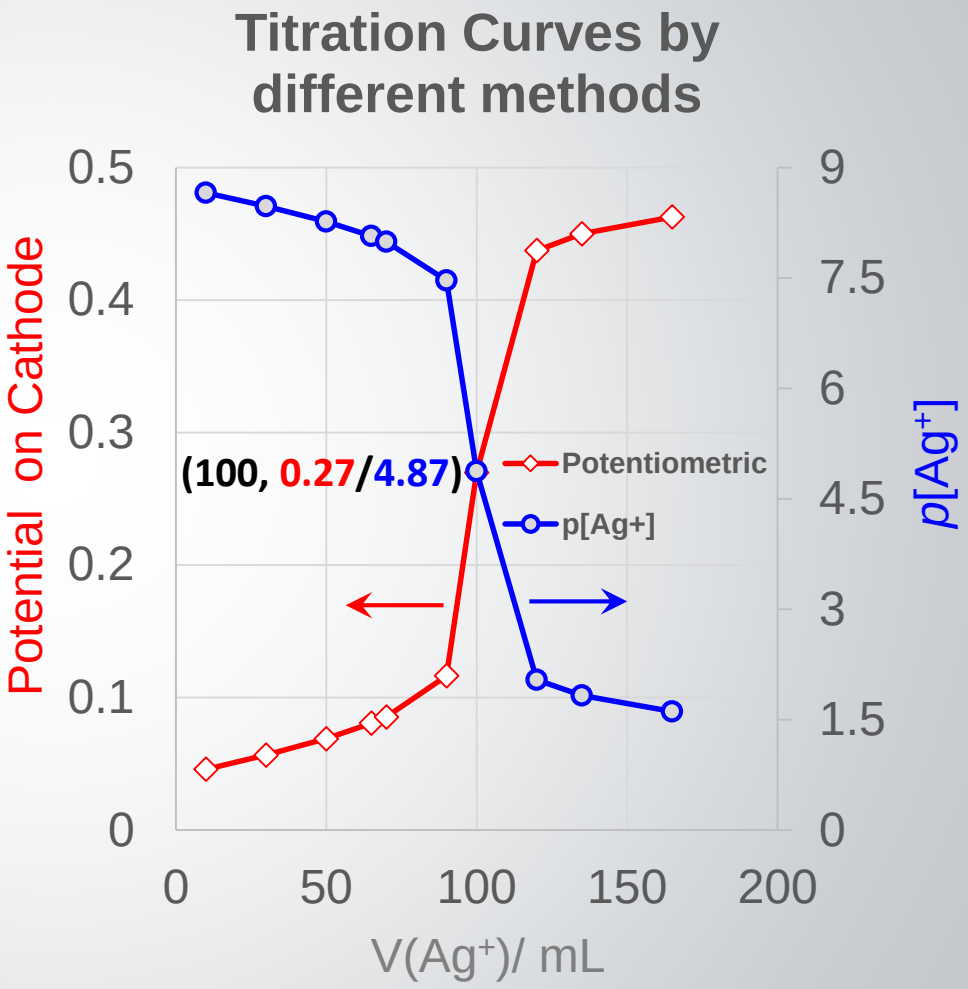
$$[Ag^+] = \frac{35\ mL \times 0.1\ mol/L}{100\ mL + 135\ mL} = 0.0149\ M$$

$$E_+ = 0.558 + 0.05916 \log [Ag^+] = 0.450\ V$$

(a) Potentiometric on precipitation titration

K_{sp} (AgCl) = 1.8×10^{-10}

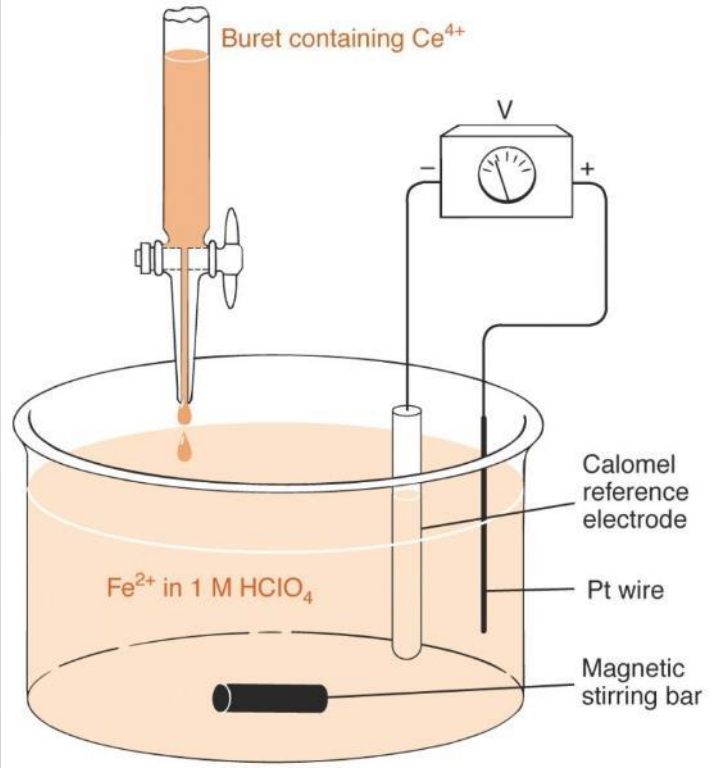


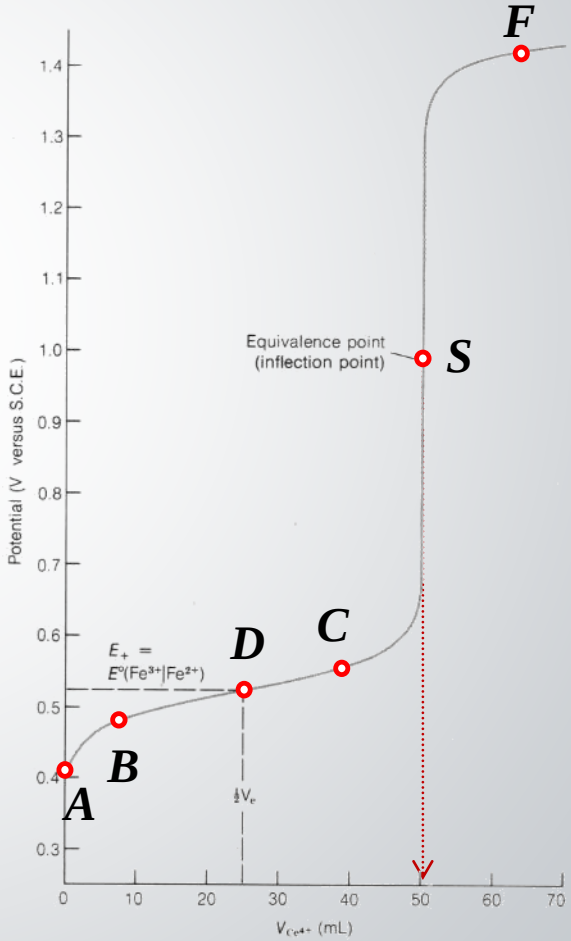


$E = 0.558 + 0.05916 \log [Ag^+]$ ⇨ $E\uparrow = 0.558 - 0.05916\, p[Ag^+]\downarrow$

(b) Potentiometric on redox - titration

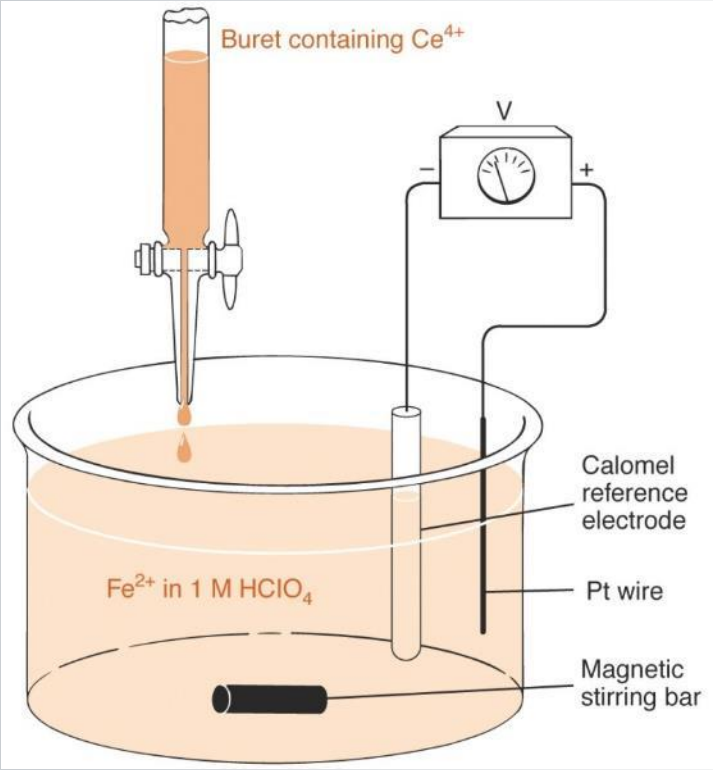
Application of indicator electrode for measuring potentials can be used for obtaining quantitative chemical information.





$$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$$

(b) Potentiometric on redox - titration



25 mL of the Fe^{2+} solution was titrated with 0.1091 M Ce^{4+} with a solvent of 1 M HClO4. The cell was monitored potentiometrically with Pt and calomel electrodes.

What is the potential during the titration at 10% of the titration,

25%, 50%, 100%, 120% and 200% ?

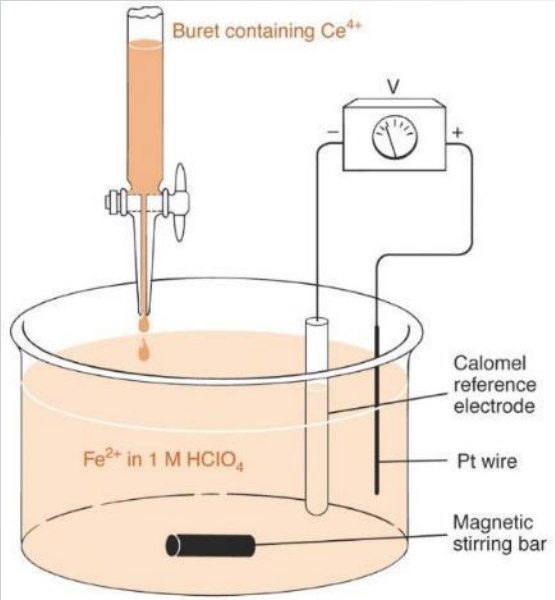
Titration reaction:

$$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$$ $K \approx 10^{16}$ in $HClO_4$

$E^{o}(Fe^{3+}/Fe^{2+}) = 0.767$ V; $E^{o}(Ce^{4+}/Ce^{3+}) = 1.70$ V; E (Ref.) = 0.241 V

(b) Potentiometric on redox - titration

At the Pt working electrode, two reactions come to equilibrium:



$$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$$

$$Fe^{3+} + e \rightleftarrows Fe^{2+} \quad E^0 = 0.767\ V$$

$$Ce^{4+} + e \rightleftarrows Ce^{3+} \quad E^0 = 1.70\ V$$

$$E = E^0{}_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

$$E = E^0{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{Ce^{3+}}{Ce^{4+}}$$

The final potential of the cell will be:

$$E(cell) = E_{+} - E_{-}(calomel_electrode)$$

*Four regions for titration curve

1. Before equivalence point, 0< V <V_e

There is almost no Ce^{4+}, so it will be better to work the potential use iron element. We know how much Ce^{4+} we add which will give how much of the $[Fe^{3+}]$. For example, **10%** titration:

$$Ce^{4+} + Fe^{2+} \leftrightarrow Ce^{3+} + Fe^{3+}$$

$$0.9 \qquad\qquad 0.1$$

$$\frac{Fe^{2+}}{Fe^{3+}} = \frac{0.9}{0.1} = 9$$

$$E = E^{0}{}_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

$$E(cell) = 0.767 - 0.05916\log(9) - 0.241 = 0.470V$$

Very tiny

Test yourself: How much of $[Ce^{4+}]$ in the solution? $[Ce^{4+}] = 1.95\times10^{-16}\%$

*Four regions for titration curve

1. Before equivalence point, 0< *V* <V_e

There is almost no Ce^{4+}, so it will be better to work the potential use iron element. We know how much Ce^{4+} we add which will give how much of the $[Fe^{3+}]$. For example, **25%** titration:

$$Ce^{4+} + Fe^{2+} \leftrightarrow Ce^{3+} + Fe^{3+}$$

$$0.75 \qquad\qquad 0.25$$

$$\frac{Fe^{2+}}{Fe^{3+}} = \frac{0.75}{0.25} = 3$$

$$E = E^{0}{}_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

$$E(cell) = 0.767 - 0.05916\log(3) - 0.241 = 0.498V$$

*Four regions for titration curve

2. $V = \frac{1}{2}V_e$

There is almost no Ce^{4+}, so it will be better to work the potential use iron element. We know how much Ce^{4+} we add which will give how much of the $[Fe^{3+}]$. For example, **50%** titration:

$$Ce^{4+} + Fe^{2+} \leftrightarrow Ce^{3+} + Fe^{3+}$$

$$0.5 \qquad\qquad 0.5$$

$$\frac{Fe^{2+}}{Fe^{3+}} = \frac{0.5}{0.5} = 1$$

$$E = E^{0}{}_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

$$E(cell) = 0.767 - 0.241 = 0.526V$$

*Four regions for titration curve

3. At equivalence point, $V = V_e$ with 100% titration

Since the reaction is in equilibrium, a tiny amount of Fe^{2+} may be formed, which will be equal as that of Ce^{4+}, so we need to use both equations. Log *a* + log *b*= log *ab*

$$Ce^{4+} + Fe^{2+} \leftrightarrow \underset{1}{Ce^{3+}} + \underset{1}{Fe^{3+}}$$

$$E = E^0{}_{Fe3+/Fe2+} - \frac{0.05916}{1}\log\frac{Fe^{2+}}{Fe^{3+}}$$

$$E = E^0{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{Ce^{3+}}{Ce^{4+}}$$

$$2E = E^0{}_{Fe3+/Fe2+} + E^0{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{\cancel{Fe^{2+}}}{\cancel{Fe^{3+}}}\cdot\frac{\cancel{Ce^{3+}}}{\cancel{Ce^{4+}}}$$

*Four regions for titration curve

3. At equivalence point, $V = V_e$ with 100% titration

Since the reaction is in equilibrium, a tiny amount of Fe^{2+} may be formed, which will be equal as that of Ce^{4+}, so we need to use both equations. Log *a* + log *b*= log *ab*

$$2E = E^0{}_{Fe3+/Fe2+} + E^0{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{\cancel{Fe^{2+}}}{\cancel{Fe^{3+}}}\cdot\frac{\cancel{Ce^{3+}}}{\cancel{Ce^{4+}}}$$

$$2E = E^0{}_{Fe3+/Fe2+} + E^0{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log(1)$$

$$E = \frac{(0.767 + 1.70)}{2} = 1.234V$$

$$E(cell) = 1.234 - 0.241 = 0.993\ V$$

So, indicator potential at equivalence-point is the average of the two standard reduction potentials of analyte and titrant, which is independent of the concentrations and volumes of the reactants.

*Four regions for titration curve

4. After equivalence point, $V > V_e$

we don't know the $[Fe^{2+}]$, which has been "used up". There are excess of the Ce^{+} and we know the quotient of $[Ce^{3+}]/[Ce^{4+}]$, e.g. **120%** titration, the value is 5.

$$Ce^{4+} + Fe^{2+} \leftrightarrow Ce^{3+} + Fe^{3+}$$

$$0.2 \qquad 1 \qquad 1$$

$$E(cell) = E^{0}{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{Ce^{3+}}{Ce^{4+}} - 0.241$$

$$E(cell) = 1.70 - \frac{0.05916}{1}\log(5) - 0.241 = 1.418V$$

*Four regions for titration curve

4. After equivalence point, $V > V_e$

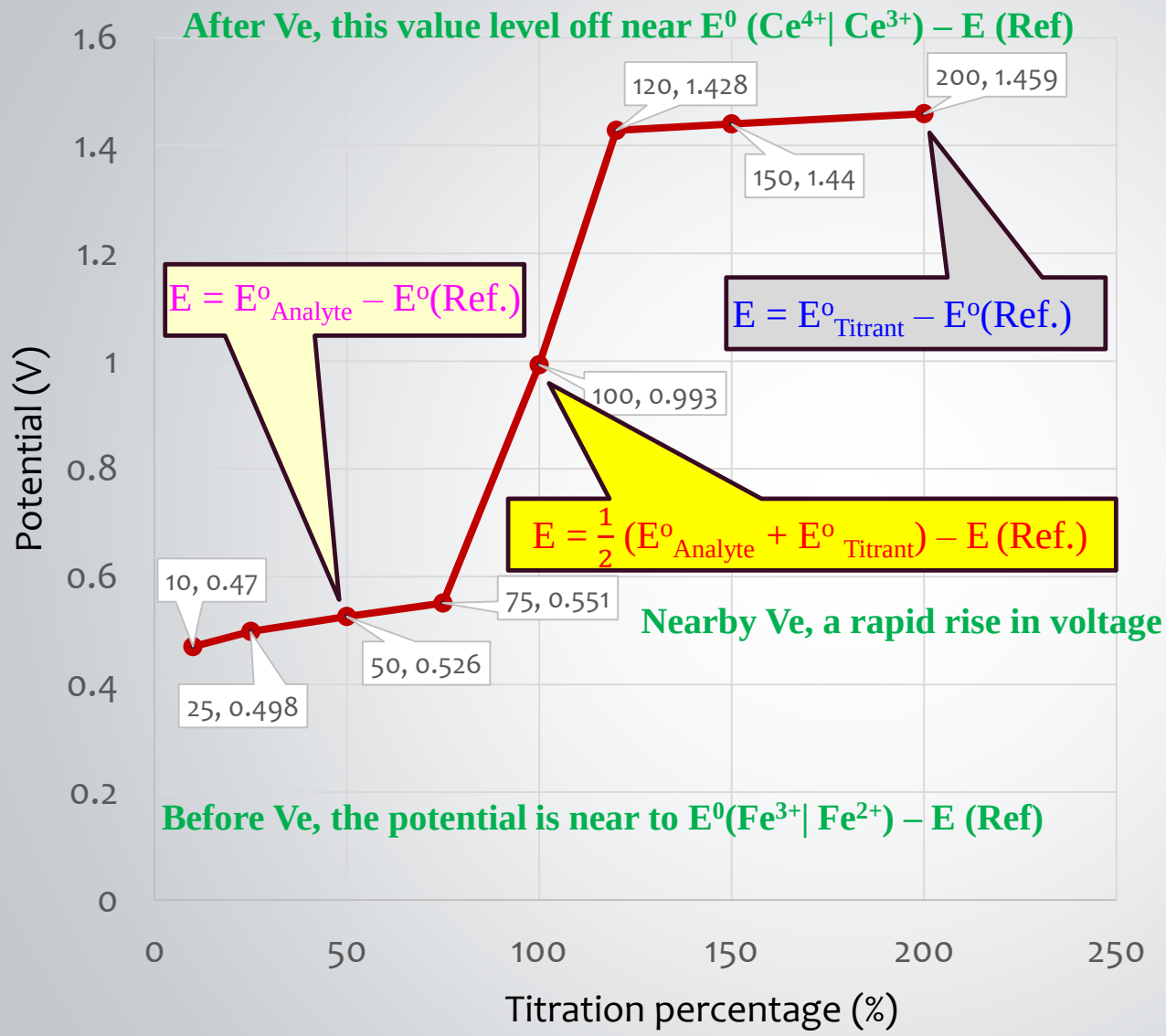
we don't know the $[Fe^{2+}]$, which has been "used up". There are excess of the Ce^{+} and we know the quotient of $[Ce^{3+}]/[Ce^{4+}]$, e.g. **200%** titration, the value is 1.

$$\underset{1}{Ce^{4+}} + Fe^{2+} \leftrightarrow \underset{1}{Ce^{3+}} + \underset{1}{Fe^{3+}}$$

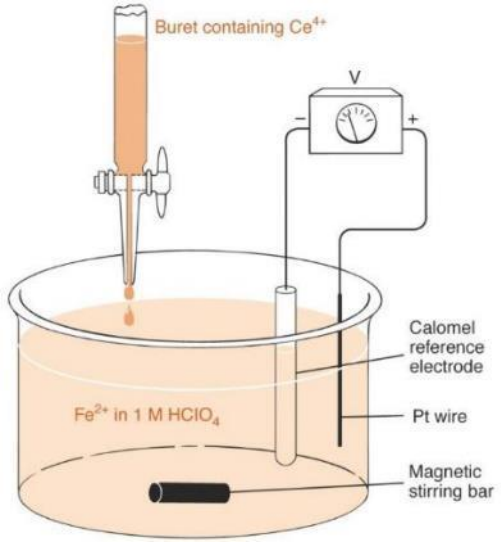
$$E(cell) = E^{0}{}_{Ce4+/Ce3+} - \frac{0.05916}{1}\log\frac{Ce^{3+}}{Ce^{4+}} - 0.241$$

$$E(cell) = 1.70 - \frac{0.05916}{1}\log(1) - 0.241 = 1.459V$$

(b) Potentiometric on redox - titration



$E^0\ (Fe^{3+}/Fe^{2+}) = 0.767\ V$
$E^0\ (Ce^{4+}/Ce^{3+}) = 1.70\ V$
$E\ (Ref.) = 0.241\ V$



$Ce^{4+} + Fe^{2+} \rightarrow Ce^{3+} + Fe^{3+}$

The larger ΔE between E°(analyte) vs. E°(titrant), the titration error will be less, the titration accuracy would be higher.

Indicator electrode

Metal indicator electrode:

- Inert electrodes made of Pt, Au, Ag and Cd etc. (transmit *e* only)

- Used for measuring potential resulting from redox reaction on the metal surface.

Ion selective electrode:

- An electrode reacting to a single ion (not redox reaction), based on a membrane of selective affinity to the ion.

- Measures the potential resulting from ions passing through the membrane